

# Software Test Plan (STP) - VOICE ASSISTANT FOR MULTIMEDIA CONTROLS

## Project Information

**Project:** Voice Assistant for MultiMedia Controls

**Document:** Software Test Plan (STP)

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## 1. Introduction

**Purpose:** This Software Test Plan (STP) defines the testing objectives, scope, approach, and activities for the Voice Assistant for MultiMedia Controls, Version 1.0. The purpose of testing is to verify that the system correctly captures and recognizes supported voice commands, executes the corresponding multimedia operations, provides appropriate feedback, and satisfies the specified functional, non-functional, security, privacy, accessibility, portability, and maintainability requirements defined in the Software Requirements Specification (SRS).

The test plan provides a structured basis for designing and executing test cases, validating system requirements, identifying defects, and verifying that the system satisfies the defined acceptance criteria.

**Scope:** Testing covers the voice-based multimedia control functions of the Voice Assistant for MultiMedia Controls, including:

- Voice command input and processing.
- Recognition of supported multimedia commands.

- Play, pause, and resume media operations.
- Volume up and volume down operations.
- Mute and unmute operations.
- Next-track and previous-track navigation.
- Execution of recognized commands on the connected media player.
- Control of multimedia playback without direct interaction with the media player interface.
- Feedback after processing a voice command.
- Handling of unrecognized or unsupported commands.
- Operation on the selected hardware platform and communication with the media player.
- Performance, reliability, security, privacy, accessibility, portability, and maintainability requirements.

Testing will cover the relevant software modules, hardware interfaces, communication interfaces, and system-level multimedia control functions defined in the SRS.

The internal implementation of third-party media players and external services beyond the interfaces required for integration are outside the scope of testing.

## **References:**

- **Voice Assistant for MultiMedia Controls — Software Requirements Specification (SRS), Version 1.0**, dated 03-08-2026.
- SRS Section 4 — System Features (Detailed).
- SRS Section 5 — Non-Functional Requirements (Detailed).
- SRS Section 5.1 — Security.
- SRS Section 6 — Quality Attributes & Acceptance Tests.
- SRS Section 8 — Requirements Traceability Matrix (RTM).

## **Definitions:**

1. SRS: Software Requirements Specification
2. STP: Software Test Plan
3. RTM: Requirements Traceability Matrix
4. FR: Functional Requirement
5. NFR: Non-Functional Requirement
6. TC: Test Case
7. MQTT: Message Queuing Telemetry Transport — communication protocol specified for possible microcontroller-to-host communication.
8. UART: Universal Asynchronous Receiver Transmitter — serial communication interface specified for communication between the microcontroller and host.
9. HUD: Head-Up Display — visual overlay showing information such as the live webcam feed, hand-landmark tracking, recognized voice commands, and media state.
10. MediaPipe: Python framework used for real-time hand-landmark detection and multi-point coordinate extraction.
11. Wake Word: A distinct trigger word or phrase that must be detected before a multimedia command is executed.

## **2. Test Items**

### **-Voice Processing and Speech Recognition**

- Processing of voice input.
- Recognition of supported voice commands.
- Wake-word/trigger detection.

### **-Command Validation and Processing**

- Validation of recognized commands.
- Mapping of recognized commands to the corresponding multimedia operations.
- Handling of unrecognized or unsupported commands.

## **-Media Control**

- Play.
- Pause.
- Resume.
- Volume Up.
- Volume Down.
- Mute/Unmute.
- Next Track.
- Previous Track.

## **-Media Player Integration**

- Execution of recognized multimedia commands on the connected media player.

## **-Feedback Module**

- Audio, visual, or textual feedback after command processing.
- Feedback for unrecognized commands.

## **-Hardware and Communication Interface**

- Operation on the selected Pi/ESP-based hardware platform.
- Communication between the hardware platform and host system.
- USB/UART or Wi-Fi-based communication where applicable.

## **-HUD and MediaPipe Interface**

- Live webcam feed.
- Hand-landmark tracking.
- Display of recognized voice commands and media state.

## **-Security and Privacy Functions**

- Local processing of audio and video.
- Wake-word/trigger requirement.

- Encrypted hardware-to-host communication where Wi-Fi communication is used.
- Protection of debug logs from raw audio/video content.
- Local-only access to the HUD/debug interface.

### **-Logging**

- Recording of command metadata, timestamps, and execution status.
- Verification that raw audio/video is not stored.

### **-Cross-Platform Compatibility**

- Operation on Windows 10/11, macOS, and Ubuntu Linux with Python 3.8+.

## **3. Features to be Tested**

Features mapped to SRS requirement IDs:

### **Voice Processing and Speech Recognition**

- FR-01: Accept voice commands through a connected microphone
- FR-12: Support a predefined set of voice commands for all multimedia operations
- NF-2: Voice command recognition accuracy  $\geq 90\%$  in ambient noise  $\leq 60$  dB

### **Command Validation and Processing**

- FR-02: Recognize commands for playing media
- FR-03: Recognize commands for pausing and resuming media playback
- FR-11: Notify the user when a voice command is not recognized
- NF-1: End-to-end latency from command to executed action  $\leq 1.5$  seconds

### **Media Control**

- FR-04: Pause currently playing media on pause command
- FR-05: Increase media volume on volume-up command
- FR-06: Decrease media volume on volume-down command
- FR-07: Mute/unmute media audio on corresponding voice command
- FR-08: Play next media item on next-track command
- FR-09: Play previous media item on previous-track command

### **Media Player Integration**

- FR-13: Execute recognized multimedia commands on the connected media player
- FR-14: Control multimedia playback without directly interacting with the media player interface

### **Feedback Module**

- FR-10: Provide feedback after processing a voice command
- FR-11: Notify the user when a voice command is not recognized

### **Hardware and Communication Interface**

- FR-15: Operate on the selected hardware platform and communicate with the media player

### **HUD and MediaPipe Interface**

- Display of live webcam feed, hand-landmark tracking, recognized voice commands, and media state (verified via UI/HUD inspection; not tied to an individual FR ID)

### **Security and Privacy Functions**

- NF-3: 100% local audio/video processing, no cloud transmission
- PRJ-SR-001: Local-only audio/video processing (no external transmission)
- PRJ-SR-002: Wake word/trigger gesture required before command execution

- PRJ-SR-003: Encrypted microcontroller-to-host communication (WPA2/TLS)
- PRJ-SR-005: HUD/debug interface accessible only locally, not network-exposed

### **Logging**

- NF-6: Local debug logging without raw audio/video storage
- PRJ-SR-004: Debug logs contain metadata only, never raw audio/video

### **Cross-Platform Compatibility**

- NF-5: Operation on Windows 10/11, macOS, and Ubuntu Linux with Python 3.8+

### **4. Features Not to be Tested**

- Internal implementation of third-party media players (e.g., Spotify, VLC, OS-native media apps) beyond the command interfaces required for integration
- Underlying MediaPipe ML SDK algorithms and hand-landmark detection model accuracy (assumed validated by the framework maintainers)
- Speech recognition engine's core model accuracy (Vosk/SpeechRecognition/Whisper) outside of the system's configured command vocabulary
- ESP32/Raspberry Pi firmware and low-level hardware driver behavior (vendor/platform responsibility)
- Operating system-level media event handling internals (Windows/macOS/Linux OS APIs)
- Physical durability, electrical safety, or hardware manufacturing quality of microcontroller boards, webcams, and microphones

## 5. Test Approach / Strategy

### Levels:

- **Unit tests (module-level):** Individual components tested in isolation — Microphone Interface, Voice Input Handler, Audio Preprocessor, Speech Recognition Module, Command Parser, Media Controller, Playlist Controller, Audio Controller, Feedback Handler, Error Handler, and Logger Module.
- **Integration tests (Voice Pipeline ↔ Media Player):** Verifying the flow from voice capture through wake-word detection, command recognition, and command parsing correctly triggers actions on the connected media player (Media Command Interface).
- **System tests (end-to-end voice-to-media-action):** Full pipeline validation — spoken command → wake word gate → recognition → execution → media player/speaker response — covering all supported commands (Play, Pause, Resume, Next, Previous, Volume Up/Down, Mute/Unmute).
- **Acceptance tests (UAT):** Stakeholder (End-User/Developer) validation against the acceptance test suites defined in Section 6: Voice Command Recognition, Playback Control, Volume Control, Track Navigation, Latency/Performance, Security & Privacy, and Accessibility.

### Types:

- **Functional testing:** Covers FR-01 through FR-15 — command capture, recognition, playback control, volume/mute control, track navigation, and feedback on unrecognized commands (test cases TC-VOICE-01 to TC-ERR-01).
- **Regression testing:** Re-running the full RTM test suite after changes to the ML pipeline, hardware actuation layer, or media

player bindings to ensure no previously passing requirement (N/P/A status) regresses.

- **Performance testing:** End-to-end latency testing per NF-1/TC-Perf-01 — 90th percentile latency  $\leq 1.5$ s across 50 test commands; also covers thermal/battery impact from continuous 30–60 FPS video capture.
- **Security & privacy testing:** Verifying 100% local processing with zero outbound audio/video streams (NF-3, PRJ-SR-001), encrypted microcontroller-host communication (PRJ-SR-003), and log content audits confirming no raw audio/video storage (PRJ-SR-004).
- **Usability/accessibility testing:** Manual walkthrough confirming audio/spoken confirmation for every executed command (NF-4/TC-UX-01), high-contrast HUD markers, and adjustable sensitivity thresholds.
- **Portability testing:** Verifying install and command execution across Windows 10/11, macOS, and Ubuntu Linux with Python 3.8+ without code changes (NF-5/TC-Port-01).

Entry Criteria: Stable build delivered, test data available, test environment ready.

Exit Criteria: 100% of planned test cases executed, 0 critical defects open, all acceptance criteria satisfied.

### 5.1 Security Validation

**Validate wake-word gating (no bypass):** Confirm the system rejects any command issued without a preceding valid wake word/trigger gesture, and that no command executes on ambient noise or accidental gesture detection alone (PRJ-SR-002 / TC-Sec-02).

**Validate local-only processing (no data exfiltration):** Network traffic capture during operation to confirm zero raw audio/video packets are

transmitted to any external or cloud endpoint (PRJ-SR-001, NF-3 / TC-Sec-01).

**TLS/WPA2 verification for microcontroller link:** Confirm the USB-Serial or Wi-Fi (MQTT/WebSockets) channel between the host and ESP32/Raspberry Pi uses at minimum WPA2 or TLS encryption, with no plaintext command transmission (PRJ-SR-003 / TC-Sec-03).

**Debug log content audit:** Inspect logger output to confirm only command metadata (timestamp, recognized command, execution status) is stored — never raw audio buffers or video frames (PRJ-SR-004 / TC-Sec-04).

**HUD/debug interface exposure check:** Network scan to confirm the live camera feed/debug UI is bound to localhost only and not reachable from other devices on the network (PRJ-SR-005 / TC-Sec-05).

**Fuzzing for voice/gesture input:** Feed malformed, garbled, or adversarial audio input and rapid/ambiguous hand-landmark sequences to the Command Parser and Validator to confirm the system fails safe (no unintended command execution) rather than crashing (FR-11, Security Objective #4).

**Fail-safe behavior on ambiguous input:** Verify that unrecognized or low-confidence commands are routed to error handling (UC-10) and never default to executing a media action.

**Multi-user/spoofing resistance:** Test with multiple people in the camera's field of view or background objects resembling hand landmarks to confirm the system doesn't misattribute gesture commands to the wrong source (Section 2.5 Constraints).

## 6. Traceability Aspects with SRS

Every requirement in SRS v1.0 (Sections 4, 5 and 5.1.2) is mapped below to the test case that verifies it. The mapping is used to confirm coverage, to scope regression when a requirement changes, and to provide the acceptance evidence required by SRS Section 6. Requirement IDs follow SRS Sections 4 and 5.

| Req ID | Requirement (short form)                                  | SRS Ref | Test Case ID(s)            |
|--------|-----------------------------------------------------------|---------|----------------------------|
| FR-01  | Accept voice commands through a connected microphone      | 4       | TC-VOICE-01                |
| FR-02  | Recognize commands for playing media                      | 4       | TC-VOICE-02,<br>TC-STT-01  |
| FR-03  | Recognize pause and resume commands                       | 4       | TC-VOICE-03,<br>TC-PLAY-03 |
| FR-04  | Pause currently playing media                             | 4       | TC-PLAY-02                 |
| FR-05  | Increase volume on volume-up command                      | 4       | TC-VOL-01                  |
| FR-06  | Decrease volume on volume-down command                    | 4       | TC-VOL-02                  |
| FR-07  | Mute and unmute media audio                               | 4       | TC-MUTE-01                 |
| FR-08  | Play next media item                                      | 4       | TC-NEXT-01                 |
| FR-09  | Play previous media item                                  | 4       | TC-PREV-01                 |
| FR-10  | Provide feedback after a command is processed             | 4       | TC-FB-01                   |
| FR-11  | Notify the user when a command is not recognized          | 4       | TC-ERR-01                  |
| FR-12  | Support the predefined voice command set                  | 4       | TC-CMD-01                  |
| FR-13  | Execute recognized commands on the connected media player | 4       | TC-PLAY-01                 |
| FR-14  | Control playback without using the media player interface | 4       | TC-PLAY-01, TC-VOL-01      |
| FR-15  | Operate on the selected hardware platform                 | 4       | TC-HW-01 (to be created)   |
| NF-1   | End-to-end latency under 1.5 s (90th percentile)          | 5       | TC-PERF-01                 |
| NF-2   | Recognition accuracy of at least 90% at noise up to 60 dB | 5       | TC-ACC-01                  |
| NF-3   | All audio and video processed locally                     | 5       | TC-SEC-01                  |
| NF-4   | Accessible feedback for every executed command            | 5       | TC-UX-01                   |
| NF-5   | Runs on Windows, macOS and Ubuntu with Python 3.8+        | 5       | TC-PORT-01                 |
| NF-6   | Local debug log with no raw audio or video stored         | 5       | TC-MAINT-01                |

| Req ID     | Requirement (short form)                          | SRS Ref | Test Case ID(s) |
|------------|---------------------------------------------------|---------|-----------------|
| PRJ-SR-001 | No raw audio or video sent to external servers    | 5.1.2   | TC-SEC-01       |
| PRJ-SR-002 | Wake word required before any command executes    | 5.1.2   | TC-SEC-02       |
| PRJ-SR-003 | Encrypted microcontroller to host link (WPA2/TLS) | 5.1.2   | TC-SEC-03       |
| PRJ-SR-004 | Debug logs store metadata only                    | 5.1.2   | TC-SEC-04       |
| PRJ-SR-005 | HUD and debug interface accessible locally only   | 5.1.2   | TC-SEC-05       |

Coverage: 25 of 26 requirements are mapped to an existing test case (96%). Acceptance requires 100% coverage with all high-priority requirements marked Verified in the SRS RTM.

Open traceability issues:

- SRS Section 4 and the RTM in SRS Section 8 use conflicting FR numbering (for example FR-07 and FR-10). The RTM must be renumbered to match Section 4.
- FR-15 has no test case in the SRS RTM. TC-HW-01 is to be created.
- The HUD and MediaPipe interface has no FR ID in the SRS, so it cannot be traced.